

FluxFraud: mathematical foundations

A stochastic transaction world, derived from the ground up

Notes for a final-year physics undergraduate

Probability, stochastic dynamics, interacting histories and controlled fraud experiments

Purpose. This document explains how a synthetic world produces customers, transaction times, recipients, amounts, fraud labels and balances. The central object is the generative process. Machine learning enters at the end as an observer that attempts to infer hidden labels from observable consequences.

The model combines heterogeneous agents, an hourly Markov chain, a mean-reverting Gaussian process, self-exciting event times, reinforced probabilistic recipient choice and a conserved ledger. These are mathematically explicit mechanisms. Their numerical parameters are design choices for a research environment, not estimates obtained from a bank's customer population.

How each explanation is organised. We specify the random variables and their units, derive the relevant relation, work through numerical values, and explain what the result means behaviourally. We distinguish a result for a simplified reference process from a statement that is actually true of the full simulator.

The examples use the current default process parameters. When a duration is required, we use a 30-day run with 64 accounts, matching the application's starting configuration. The command-line configuration instead starts with 160 accounts and 45 days; the mathematical mechanisms are the same.

Reading prerequisites. Calculus, linear algebra, elementary probability and familiarity with differential equations are sufficient. Stochastic calculus is introduced only as far as the mean-reversion derivation requires. There are no exercises. A separate implementation companion locates these equations in the project and explains the execution sequence.

Scientific scope. A reproducible synthetic benchmark can establish how a detector behaves in the world we specified. It cannot, by itself, establish performance on real financial transactions. Citations identify the origins of mathematical tools; they do not validate our chosen fraud rates or customer assumptions.

18 September 2026

Reading map and notation

The logical flow. Pages 3-6 build a population from probability distributions. Pages 7-13 give customers evolving hidden states. Pages 14-19 generate event times. Pages 20-25 attach recipients and transaction attributes. Pages 26-30 turn these events into a temporal network and ledger. Pages 31-35 explain changing environments, Monte Carlo experiments and verification. Pages 36-39 connect the simulated world to causal detection experiments and establish its limitations. References occupy pages 40-41.

Time and money. Time t is measured in hours; h is an integer hour. The horizon is $H = 24D$ for D days. Money is recorded as integer minor units, with 100 minor units per major currency unit. We write $L = 1$ major currency unit when defining dimensionless logarithms. An amount of 45 therefore means 45L, not an unspecified physical dimension inside a logarithm.

Accounts and hidden variables. There are N accounts, indexed by i and j . Each account has a community g , baseline log-spending level m with superscript 0, and activity multiplier a . Its evolving state consists of behaviour S , compromise indicator C and current log-spending level x . The hidden set R contains preferred recipients for fraudulent events.

Events and history. N with a subscript i and argument t counts sender i 's attempted transactions. Λ is its event intensity. We use z for residual self-excitation, n with subscripts ij for previous attempts from i to j , B for balance, A for attempted amount, and Y for the true fraud label. The same letter N denotes population size only when it has no time argument or account subscript.

$$H_t^- = \{\text{events strictly before } t\}, \quad Y \in \{0, 1\}. \quad (2.1)$$

Conditioning matters. A statement about an amount given one customer's present x differs from a statement about all customers pooled together. Similarly, a long-run result with a fixed spending target differs from the distribution under switching targets. Most apparent contradictions disappear when the conditioning information is stated.

References. Underlined source names are clickable. Research papers provide provenance; linked university and NIST materials provide entry points to general concepts. All numerical worked examples are calculations for this simulator, rather than values reported by those papers.

1. What is random, and what is fixed?

A simulation specifies a probability law and then draws one realisation from it. If a coin has probability p of heads, the parameter p is fixed while the outcome is random. Here a customer's activity is itself drawn at initialisation and then held fixed throughout that run. Their transactions remain random even after this activity has been fixed.

For a continuous variable X , the probability of an interval is an integral of a density. A density is not a probability at a single point and can exceed one. For a discrete state, probabilities are masses that sum to one. These two types coexist: transaction times are continuous, while behavioural states are discrete.

$$\Pr(a < X < b) = \int_a^b f_X(x) dx, \quad \sum_s \Pr(S = s) = 1. \quad (3.1)$$

The Gaussian distribution supplies the basic additive fluctuation. Its parameters are a location m and standard deviation s ; the second argument of the notation below is the variance. Z denotes a standard Gaussian draw.

$$X = m + sZ, \quad Z \sim \mathcal{N}(0, 1), \quad f_X(x) = \frac{e^{-(x-m)^2/(2s^2)}}{s\sqrt{2\pi}}. \quad (3.2)$$

Concrete example. If X has mean 3 and standard deviation 0.6, a draw $Z = 1$ gives $X = 3.6$, while $Z = -1$ gives $X = 2.4$. Approximately 68.3% of Gaussian draws lie within one standard deviation of the mean. A realised sequence need not contain exactly this fraction, especially when it is short.

Behavioural meaning. Randomness models variability around a tendency, rather than a hidden instruction that every customer must follow. Two identical parameter sets can produce different event histories. A pseudorandom seed makes one such history repeatable; it does not remove the stochastic interpretation.

Three conceptual layers will recur: population heterogeneity between accounts; temporal variation within an account; and event-level variation around its current condition. Confusing these layers leads to incorrect claims about spending limits, fraud prevalence and uncertainty. In particular, a broad pooled distribution can arise even when each individual account has relatively modest fluctuations.

2. Spending scale: what it actually means

The state x is a logarithmic location parameter. Exponentiating it gives the scale of ordinary attempted payments, conditional on the customer's current state. Before cent rounding, an ordinary payment has the form below, where epsilon is independently redrawn for each attempt.

$$A = L \exp(x + \varepsilon), \quad \varepsilon \sim \mathcal{N}(0, 0.60^2). \quad (4.1)$$

Taking logarithms turns multiplicative variation into additive variation. Because a centred Gaussian has median zero, half its draws are below zero and half above. Exponentiation preserves order, so the median amount is $L \exp(x)$. The mean differs because the exponential is convex.

$$\text{median}(A \mid x) = L e^x, \quad \mathbb{E}[A \mid x] = L e^{x + 0.60^2/2}. \quad (4.2)$$

A and B. Suppose A has $x = \ln 45$ and B has $x = \ln 90$. Their median ordinary attempts are 45 and 90 currency units. Their mean attempts are approximately 53.87 and 107.75. With the same illustrative fluctuation $\varepsilon = 0.30$, their attempts are 60.74 and 121.49. B's distribution is a factor of two larger at every matching quantile.

This does not mean B always spends more. If A draws $\varepsilon = 0.60$ while B draws $\varepsilon = -0.60$, A attempts about 82.00 and B about 49.39. Both distributions have unbounded positive support before rounding, so neither scale is a maximum permitted payment.

Capacity versus tendency. A higher scale means a tendency towards larger attempted payments. The ability to settle a particular attempt depends on the current balance. A customer with a 90-unit scale and a 20-unit balance can attempt a large payment and be declined. A customer with a 45-unit scale and a 5,000-unit balance can occasionally make a very large successful payment.

This separation is intentional: spending preference and available funds are different state variables. The model checks the balance after sampling the attempt. It does not resample the amount until it fits the balance.

General concept: [NIST: lognormal distributions](#). The scale interpretation above follows directly from the simulator's amount equation.

3. Building a heterogeneous population

Each account starts with its own baseline log-spending level. This is a population draw, not the event-level noise in the previous section. Community membership is also drawn once, uniformly from four groups; these groups affect recipient preferences rather than assigning income classes.

$$m_i^0 \sim \mathcal{N}(\ln 45, 0.65^2), \quad M_i = Le^{m_i^0}. \quad (5.1)$$

The baseline median scales M are lognormally distributed across customers. Their population median is 45, while their mean is about 55.58. This distinction remains even before customers change behaviour or generate individual payment noise. A normal distribution in log space becomes asymmetric in money space.

To see why, transform a variable using $x = \ln(a/L)$. The derivative is $dx/da = 1/a$. Probability mass is preserved under the change of variable, producing the Jacobian factor in the density.

$$f_M(a) = \frac{1}{0.65a\sqrt{2\pi}} \exp\left[-\frac{(\ln(a/L) - \ln 45)^2}{2(0.65)^2}\right], \quad a > 0. \quad (5.2)$$

Concrete example. Baseline log levels one population standard deviation below and above $\ln 45$ correspond to scales $45 \exp(-0.65) = 23.49$ and $45 \exp(0.65) = 86.20$. Approximately 68.3% of baseline scales lie in that interval. The central 95% interval is approximately 12.59 to 160.89, using 1.96 standard deviations. These are population intervals, not limits for any one account's payments.

Behavioural meaning. Some customers habitually attempt smaller payments and others larger ones. The Gaussian assumption says that multiplicative differences, rather than additive differences in currency, are the natural measure of heterogeneity. Doubling from 20 to 40 is the same log change as doubling from 100 to 200.

The value 45 and dispersion 0.65 are modelling choices. They have not been fitted to Revolut or another institution. A real calibration would estimate an account-level distribution while distinguishing genuine heterogeneity from noisy estimates based on a few observed payments.

Distribution definitions and parameter conventions: [NIST: lognormal distributions](#).

4. Activity and balances are separate quantities

The activity multiplier controls how often an account attempts transactions. It does not directly control their sizes. It is drawn independently of the baseline spending parameter in the model's population specification.

$$\ln a_i \sim \mathcal{N}(-0.55^2/2, 0.55^2). \quad (6.1)$$

For a Gaussian Z , completing the square in the density gives $E[\exp(sZ)] = \exp(s^2 \text{ squared divided by two})$. Applying that identity to a gives the following moments. The negative location is chosen specifically to normalise the mean to one.

$$E[a_i] = 1, \quad \text{median}(a_i) = e^{-0.15125} \simeq 0.860, \quad (6.2)$$

$$\text{Var}(a_i) = e^{0.3025} - 1 \simeq 0.353. \quad (6.3)$$

Concrete example. Before daily and behavioural factors, an account with $a = 2$ has background rate 0.20 attempts per hour, while one with $a = 0.5$ has rate 0.05. Ignoring excitation and other changes, that corresponds to 4.8 versus 1.2 expected daily attempts. These are averages over possible histories, not an instruction to generate fractional transactions.

Opening balances are independent discrete uniform draws from 200,000 through 799,999 minor units. Their mean is 499,999.5 minor units, or 4,999.995 major units. The tiny half-cent in the expectation is legitimate: an average need not be an attainable individual value.

Behavioural meaning. A high-activity account need not be wealthy and need not make large payments. Four combinations are possible: frequent small payments, frequent large payments, infrequent small payments and infrequent large payments. All can start with similar balances. Subsequent balances depend on successful incoming and outgoing transfers.

The simulation therefore separates three questions: how often a customer acts, how large their attempts tend to be, and whether they currently have enough money. These components become statistically dependent later through the realised history, even though their initial draws are independent.

Lognormal moment conventions: [NIST: lognormal distributions](#). Activity-based temporal network models offer a related modelling perspective: [Perra et al. \(2012\)](#); their particular network dynamics are not used verbatim here.

5. Behaviour as an hourly Markov chain

An account occupies one of three behavioural states: ordinary, travel or high spending, encoded as 0, 1 and 2. Every hour, the next state depends on the current state through a transition matrix. This is the Markov property for this component; it does not imply that the entire observable transaction history has no memory.

$$\Pr(S_{h+1} = b \mid S_h = a, S_{h-1}, \dots) = P_{ab}. \quad (7.1)$$

The rows below are ordered ordinary, travel and high spending. Each row sums to one. The row is selected using the account's old state, then one new state is sampled.

$$P_{0\cdot} = (0.990, 0.006, 0.004), \quad (7.2)$$

$$P_{1\cdot} = (0.035, 0.960, 0.005), \quad P_{2\cdot} = (0.080, 0.010, 0.910). \quad (7.3)$$

Concrete example. An ordinary customer has a 99% chance of remaining ordinary for the next hour, a 0.6% chance of entering travel and a 0.4% chance of entering high spending. Using a uniform draw U between zero and one, the intervals $[0,0.990)$, $[0.990,0.996)$ and $[0.996,1)$ select these outcomes. Thus $U = 0.994$ means travel.

Behavioural meaning. Persistence creates episodes rather than independently assigning a new lifestyle at every transaction. Once travelling, a customer is likely to remain travelling for several hours. Their payment frequency, spending target and foreign-payment propensity then change together. This creates correlated observable features without making them deterministic fraud signals.

All accounts begin ordinary. At the beginning of the first simulated hour, the chain is already updated once before events are generated. The first hour is consequently almost entirely ordinary, but not necessarily exclusively so.

This transition matrix is an assumption. A state named travel is a compact latent regime, not an itinerary or geographic movement model. The numerical transitions should be interpreted as parameters of an experiment rather than measured customer behaviour.

Mathematical background: [Aldous & Fill: finite Markov chains](#).

6. Episode lengths and equilibrium occupancy

If a chain remains in state s with probability P with subscript ss at every step, its consecutive episode length K is geometric. We count the current occupied hour as the first hour, so K takes values 1, 2 and so on.

$$\Pr(K = k) = P_{ss}^{k-1}(1 - P_{ss}), \quad \mathbb{E}[K] = \frac{1}{1 - P_{ss}}. \quad (8.1)$$

The expected ordinary, travel and high-spending episodes are therefore 100, 25 and 11.11 hours. A 25-hour expected travel episode is not a fixed 25-hour duration. For example, the probability of remaining in travel through another 24 transitions is 0.96 to the power 24, approximately 0.375.

For a time-homogeneous irreducible finite chain, a stationary occupancy vector π solves the balance equations. Stationary occupancy is not obtained by simply normalising the three mean episode lengths, because transitions connect states with unequal probabilities.

$$\pi P = \pi, \quad \sum_s \pi_s = 1, \quad (8.2)$$

$$\pi \simeq (0.82176, 0.13426, 0.04398). \quad (8.3)$$

Concrete example. At stationarity, a population of 1,000 independent accounts would have about 822 ordinary, 134 travelling and 44 high-spending accounts on average at an arbitrary hour. A particular hour fluctuates around these expectations. A 64-account simulation cannot reproduce those proportions exactly, and its initial condition is not stationary.

Behavioural meaning. A state can be persistent yet uncommon because entry is rare. Conversely, frequent short episodes can occupy a substantial fraction of time. The stationary vector combines entry, exit and communication between all states.

The simulator starts with every account ordinary. Its distribution after k updates is the initial row vector multiplied by P to the power k . Stationary occupancy is thus a useful long-run reference, not the exact composition on Day 1. The amount process has its own relaxation time, so an account entering travel does not instantly acquire the full travel spending scale.

Stationary distributions and finite-chain assumptions: [Aldous & Fill: finite Markov chains](#).

7. Compromise, recovery and true fraud

Compromise is a separate binary state C . An ordinary or travelling customer can be compromised; the behavioural state and compromise state represent different mechanisms. With $q = 0.0007$ per hour and $r = 0.10$ per hour, the simulator converts hazard parameters into hourly transition probabilities.

$$u(d) = 1 - e^{-q(1+d)}, \quad v = 1 - e^{-r}, \quad (9.1)$$

$$\Pr(C_{h+1} = 1 \mid C_h = 0) = u(d), \quad \Pr(C_{h+1} = 0 \mid C_h = 1) = v. \quad (9.2)$$

Before the shift, u is about 0.00069976 and v about 0.095163. The mean compromised episode lasts $1/v = 10.51$ occupied hours. At fixed d , this two-state discrete chain has compromised occupancy $u/(u+v)$, approximately 0.00730 before the shift. This is an account-time fraction, not a transaction fraud rate.

Concrete example. Among 10,000 currently uncompromised accounts, about seven enter compromise in the next update on average. Among 100 compromised accounts, about 9.52 recover. A newly compromised account is not also recovered within the same update: the old state determines which transition applies.

At a transaction event, the true fraud indicator is sampled using the sender's compromise state.

$$Y = C_i V, \quad V \sim \text{Bernoulli}(0.65). \quad (9.3)$$

A compromised customer can therefore produce legitimate-labelled attempts as well as fraudulent ones. Fraud labels are known to the simulator because it sampled them; a detector does not observe C or Y when making its prediction.

A subtle discretisation point. The hazard-to-probability conversion gives the probability of at least one event from a constant hazard over one hour. The implemented two-state update is not the exact transition matrix of a continuous-time infection-recovery chain allowing multiple jumps within that hour. Such a chain would use a matrix exponential and would have transition probability $q/(q+r)$ times $[1 - \exp(-(q+r))]$ from safe to compromised. Our hourly model is a deliberate discrete specification.

Exponential waiting-time convention: [NIST: exponential distributions](#); discrete-chain reasoning: [Aldous & Fill: finite Markov chains](#).

8. Why introduce Brownian fluctuations?

Customer spending tendencies should vary gradually instead of being independently redrawn at each event. A continuous-time reference model represents many small unobserved influences by Brownian noise. Brownian increments over disjoint intervals are independent Gaussian variables with variance equal to elapsed time.

$$W_{t+\Delta} - W_t \sim \mathcal{N}(0, \Delta), \quad \mathbb{E}[dW_t] = 0, \quad (dW_t)^2 = dt. \quad (10.1)$$

The last relation is an Itô-calculus rule describing quadratic variation, not an equality between ordinary infinitesimals. Noise accumulated over a short interval is of order the square root of that interval. This is why simply adding a fixed-size shock at every smaller numerical step would change the model.

For log-spending x , an unconstrained Brownian motion would have variance increasing indefinitely. Mean reversion supplies a restoring tendency towards a target m , analogous to overdamped motion in a harmonic potential.

$$dx_t = \kappa(m - x_t) dt + \sigma dW_t, \quad U(x) = \frac{\kappa}{2}(x - m)^2. \quad (10.2)$$

Here kappa has units inverse hours, sigma has units inverse square-root hours because x is dimensionless, and U is a mathematical potential rather than money or thermodynamic energy. The deterministic drift is minus the derivative of U .

Concrete example. If x is 0.5 above its target, $\kappa = 0.06$ produces an initial deterministic drift of -0.03 per hour. Over a sufficiently short 0.01-hour interval, the deterministic change is about -0.0003, while the noise standard deviation is 0.12 times the square root of 0.01, or 0.012. Individual short steps can move away from the target despite the restoring drift.

Behavioural meaning. Mean reversion is a tendency across possible histories, not a guarantee that tomorrow's spending scale is closer to baseline. Random fluctuations and changes in the target can temporarily dominate it. The actual simulator samples an exact one-hour transition of this reference process and holds the resulting x fixed during that hour.

Historical process: [Uhlenbeck & Ornstein \(1930\)](#). General physics explanation: [MIT: Brownian motion and mean reversion](#).

9. Deriving the exact OU transition

Within an interval on which m is fixed, define $y = x - m$. Multiplying the stochastic differential equation by $\exp(\kappa t)$ removes the restoring drift. Since this integrating factor is deterministic, the stochastic product rule produces no additional cross-variation term.

$$d(e^{\kappa t} y_t) = \sigma e^{\kappa t} dW_t. \quad (11.1)$$

Integrate from t to $t + \Delta$ and divide by the final integrating factor. The result separates memory of the initial condition from a weighted sum of subsequent noise increments.

$$x_{t+\Delta} = m + (x_t - m)e^{-\kappa\Delta} + \sigma \int_t^{t+\Delta} e^{-\kappa(t+\Delta-s)} dW_s. \quad (11.2)$$

The stochastic integral has mean zero. Its variance follows from the Itô isometry: for a deterministic integrand f , the variance of its Brownian integral is the integral of f squared. Thus the transition is Gaussian, and it can be sampled using a single standard Gaussian Z .

$$x_{t+\Delta} = m + (x_t - m)e^{-\kappa\Delta} + \sigma \sqrt{\frac{1 - e^{-2\kappa\Delta}}{2\kappa}} Z. \quad (11.3)$$

Concrete example. With $\kappa = 0.06$, $\sigma = 0.12$ and $\Delta = 1$ hour, the memory factor is 0.941765 and the innovation standard deviation is 0.116488. If x starts exactly at m and $Z = 1$, the next x is $m + 0.116488$. Its spending scale is multiplied by $\exp(0.116488)$, approximately 1.1235.

Why exact matters. An Euler step would replace the memory factor by $1 - \kappa\Delta$ and the innovation standard deviation by σ times the square root of Δ . These are small-step approximations. The simulator uses the exact fixed-target transition above, avoiding that particular time-step bias.

Exactness has a boundary: it concerns the transition conditional on a fixed target during the step. The hourly update of the behavioural target and the decision to hold x constant during event generation remain separate modelling choices. There is no continuously evaluated OU path between transactions in the current simulation.

Process foundation: [Uhlenbeck & Ornstein \(1930\)](#); pedagogical treatment: [MIT: Brownian motion and mean reversion](#).

10. OU relaxation, variance and correlation

The conditional mean relaxes exponentially towards m . Setting the initial displacement's remaining fraction to one half gives the half-life. With default κ , the relaxation time is 16.67 hours and the half-life is 11.55 hours.

$$\tau_x = \kappa^{-1}, \quad t_{1/2} = \frac{\ln 2}{\kappa}. \quad (12.1)$$

Under a fixed target and stationary initialisation, the variance and autocovariance are simple. They describe the reference process, not the full process with switching behaviour and a later shift.

$$\text{Var}(x) = \frac{\sigma^2}{2\kappa} = 0.12, \quad \text{Cov}(x_t, x_{t+s}) = 0.12e^{-\kappa|s|}. \quad (12.2)$$

Concrete example. The stationary log-scale standard deviation is the square root of 0.12, approximately 0.3464. A one-standard-deviation upward movement multiplies the spending scale by about 1.414. Log-scale correlation is approximately 0.698 after six hours and 0.237 after 24 hours. Neither figure is the correlation of raw payment amounts, which include fresh event noise.

The probability density of the continuous reference process obeys a Fokker-Planck equation: drift transports probability towards the target while diffusion spreads it out.

$$\partial_t p = \kappa \partial_x [(x - m)p] + \frac{\sigma^2}{2} \partial_x^2 p. \quad (12.3)$$

Setting the probability current to zero yields a stationary density proportional to $\exp[-\kappa(x-m)^2 / \sigma^2]$, recovering the Gaussian variance above. This is the same restoring-force versus diffusion balance familiar from elementary statistical physics.

Behavioural meaning. Increasing σ makes customer tendencies noisier. Increasing κ makes deviations fade faster and, with σ fixed, reduces the stationary variance. Changing both can preserve the marginal variance while changing the temporal correlation. A histogram alone cannot distinguish such parameter pairs; a time series can.

The simulator starts x at its baseline rather than drawing it from this stationary Gaussian. Early variation is therefore smaller than the fixed-target stationary reference would suggest, before switching targets and population heterogeneity are added.

Foundational context: [Uhlenbeck & Ornstein \(1930\)](#). The displayed moments follow from the transition on page 11.

11. Switching targets and lagged behaviour

The target depends on the customer's current behavioural state and the environmental shift d . The offsets below change the preferred log-spending level, whereas the OU process controls how quickly x follows it.

$$m_i(h) = m_i^0 + b_{S_i(h)} + 0.20d(h), \quad (b_0, b_1, b_2) = (0, 0.25, 0.65). \quad (13.1)$$

If a regime were held long enough and noise averaged away, travel would multiply the target scale by $\exp(0.25) = 1.284$ and high spending by $\exp(0.65) = 1.916$. These are target multipliers, not immediate jumps in x .

Concrete example. Take a customer starting at $x = \ln 45$ who enters high spending. Ignore noise temporarily. After one hourly update, the increase in x is $0.65[1 - \exp(-0.06)] = 0.03785$, giving a scale of about 46.74. If the state persists for ten hours, the increase is $0.65[1 - \exp(-0.60)] = 0.29327$ and the scale becomes about 60.34. The eventual fixed target would be 86.20.

For a deterministic sequence of hourly targets, repeated substitution shows how past targets are weighted. Let $a = \exp(-\kappa)$ and let ξ denote independent innovations with the variance derived earlier.

$$x_h = a^h x_0 + (1 - a) \sum_{r=1}^h a^{h-r} m_r + \sum_{r=1}^h a^{h-r} \xi_r. \quad (13.2)$$

Behavioural meaning. Recent regimes matter more than distant ones, but a customer who has just returned from a high-spending episode can retain an elevated scale. This produces a lag between the discrete behaviour label and continuous spending tendency. A transaction that looks unusually large may therefore be a legitimate consequence of recent history.

The full process is a switching linear Gaussian system when conditioned on the regime path.

Marginalising over random paths gives a mixture, not one universal Gaussian equilibrium. Quoting the fixed-target stationary variance as the total population variance would omit baseline heterogeneity, changing targets and transient effects.

The hourly convention matters: the new regime is selected, the exact one-hour OU transition is applied towards its target, and that updated value supplies all event marks in the hour. This defines the model's clock ordering.

12. The background transaction clock

Before self-excitation is added, a customer's transaction intensity combines activity, daily rhythm, behaviour and compromise. The daily multiplier is evaluated once per integer hour, so the actual background is piecewise constant even though its defining curve is a cosine.

$$s(h) = 1 + 0.55 \cos \left[\frac{2\pi((h \bmod 24) - 15)}{24} \right]. \quad (14.1)$$

It reaches 1.55 at 15:00 and 0.45 at 03:00. Its average over the 24 integer hours is exactly one. Consequently the daily rhythm redistributes a constant ordinary account's expected baseline activity across the day without changing its daily total.

$$\mu_i(h) = 0.10 a_i s(h) c_{S_i(h)} k_{C_i(h)}(d), \quad (14.2)$$

$$(c_0, c_1, c_2) = (1, 1.4, 1.8), \quad k_0(d) = 1, \quad k_1(d) = \frac{3}{1+0.5d}. \quad (14.3)$$

Concrete example. An ordinary, uncompromised customer with $a = 1$ has background rate 0.155 per hour at 15:00 and 0.045 at 03:00. At 15:00, a travelling compromised account with $a = 2$ and no shift has background rate 0.10 times 2 times 1.55 times 1.4 times 3 = 1.302 attempts per hour.

Behavioural meaning. A rate is expected event count per unit time locally, not a probability. A rate of 1.302 is perfectly valid; a probability of 1.302 would not be. For a sufficiently short interval Delta, the probability of one event is approximately rate times Delta. For a full hour the exact count law also depends on self-excitation and evolving history.

Compromise increases attempted activity, but the model does not suppress attempts when balances become low. Payment settlement is checked later. A receiver's incoming payment does not excite its own outgoing event clock. The network and ledger therefore influence recipients and settlement, while sender event rates depend on the latent state and that sender's own previous attempts.

All four multipliers are synthetic assumptions. The cosine is a controlled seasonal pattern, not an inferred schedule for any particular customer population.

13. Counting processes and conditional intensity

For sender i , the counting process records how many attempts have occurred up to time t . It increases by one at each event and is otherwise constant. Its conditional intensity describes the short-time event tendency given the relevant history and latent state.

$$N_i(t) = \sum_k \mathbf{1}\{t_{ik} \leq t\}, \quad (15.1)$$

$$\Pr(N_i(t+dt) - N_i(t) = 1 \mid \mathcal{F}_t^-) = \lambda_i(t)dt + o(dt). \quad (15.2)$$

The information set here belongs to the generator: it includes the hidden hourly state used to construct the rate. A detector observes a smaller information set. Conditioning on different information generally gives different predictive intensities.

For a homogeneous Poisson reference process with constant rate μ , independent increments yield a Poisson count over duration T and an exponential waiting time. This is the special case with no self-excitation and no changing background.

$$\Pr(N(T) = k) = e^{-\mu T} \frac{(\mu T)^k}{k!}, \quad \Pr(\tau > t) = e^{-\mu t}. \quad (15.3)$$

Concrete example. With $\mu = 0.10$ per hour, the mean waiting time is ten hours. The probability of no events during a full ten-hour interval is $\exp(-1)$, about 0.368. A ten-hour mean does not imply a regular ten-hour spacing. Clusters and long gaps occur even in a Poisson sequence by chance.

Behavioural meaning. A stochastic clock converts a tendency to transact into actual irregular event times. Self-excitation will add structured clustering on top of ordinary Poisson variability. Comparing those two sources of burstiness is more informative than assuming that every cluster indicates fraud.

For a deterministic time-varying rate, the no-event probability involves the integrated rate. With history-dependent rates, the same survival construction uses the conditional rate along the path on which no new event has yet occurred. This distinction is the foundation of the thinning algorithm on pages 18-19.

General waiting-time distribution: [NIST: exponential distributions](#). Conditional-intensity framework and financial applications: [Bacry, Mastromatteo & Muzy \(2015\)](#).

14. Hawkes self-excitation

A previous attempt temporarily increases the same sender's tendency to make another attempt. The kernel has an exponentially decaying shape, allowing the complete past to be summarised by one residual excitation variable.

$$\lambda_i(t) = \mu_i(\lfloor t \rfloor) + z_i(t), \quad z_i(t) = \sum_{t_{ik} < t} \eta \beta e^{-\beta(t - t_{ik})}. \quad (16.1)$$

The parameters are $\eta = 0.35$ and $\beta = 1.5$ per hour. An event adds $\eta \beta = 0.525$ per hour to the intensity immediately after it occurs. Between events, the residual decays exponentially. Its decay half-life is $\ln 2$ divided by β , approximately 0.462 hours or 27.7 minutes.

$$dz_i(t) = -\beta z_i(t) dt + \eta \beta dN_i(t). \quad (16.2)$$

Concrete example. For background 0.10 and zero initial excitation, an event raises the subsequent intensity from 0.10 to 0.625 per hour. If no other event occurs, the intensity after half an hour is $0.10 + 0.525 \exp(-0.75) = 0.3480$. After one hour it is about 0.2171. A second event would add another 0.525 to whatever residual remains.

Behavioural meaning. One payment can be part of an episode: paying several bills, making related purchases, or a burst during compromise. The same excitation mechanism applies to legitimate and fraudulent attempts. Bursty behaviour is therefore informative only in context.

The excitation counts attempted transactions, including those later declined by the ledger. Rejection of an internal thinning proposal, discussed next, is different: it creates no attempt and no excitation. Residual excitation also crosses hour boundaries; resetting it every hour would artificially break behavioural memory.

There is no cross-account triggering kernel in this model. Receiving a transfer does not directly raise the receiver's event rate. Nor does the amount of a payment change its excitation jump. These are possible extensions, but they are not part of the present equations.

Process family and notation: [Bacry, Mastromatteo & Muzy \(2015\)](#). The numerical example is specific to FluxFraud.

15. Branching, stability and expected counts

The area under a Hawkes kernel is its branching ratio. In the cluster interpretation, a background event is an immigrant, and events can generate descendants. The integral of the exponential kernel gives eta, independent of beta.

$$\int_0^\infty \eta \beta e^{-\beta s} ds = \eta. \quad (17.1)$$

For a constant background and subcritical eta below one, a single immigrant has expected total cluster size $1 + \eta + \eta^2 + \dots$ and so on, including itself. This geometric series explains the stationary mean rate.

$$\mathbb{E}[K_{\text{cluster}}] = \frac{1}{1-\eta}, \quad \bar{\lambda} = \frac{\mu}{1-\eta}. \quad (17.2)$$

Concrete example. At $\eta = 0.35$, the mean cluster size is 1.5385. With fixed $\mu = 0.10$, the stationary mean event rate is 0.15385 per hour, or 3.692 per day. Setting eta to zero gives 2.4 per day. Removing excitation therefore changes event volume as well as clustering unless the baseline is separately adjusted.

The simulator begins with zero excitation. Even under a truly constant background, its mean rate initially rises towards the stationary reference. Taking expectations in the residual equation gives a linear relaxation equation with solution below.

$$\mathbb{E}[\lambda(t)] = \frac{\mu}{1-\eta} [1 - \eta e^{-\beta(1-\eta)t}] \quad (z(0) = 0). \quad (17.3)$$

The relaxation time of this mean is $1/[\beta(1-\eta)]$, about 1.026 hours. It differs from the single-kernel decay time $1/\beta = 0.667$ hours because descendants can create further descendants.

Behavioural meaning. Eta controls amplification; beta controls how concentrated the immediate response is in time. Near $\eta = 1$, clusters can become large and variable. Subcritical excitation does not make the full simulator stationary: daily rhythms, Markov states, reinforced recipients and the explicit shift still change its distribution.

The cluster representation is a mathematical interpretation of the event law. The implementation generates times by thinning and does not record a genealogical parent for each payment.

Foundational cluster representation: [Hawkes & Oakes \(1974\)](#). Stationary formulas here are restricted to the stated constant-background reference.

16. Thinning: generating the event times

Within an hour, the background μ is constant. Starting from residual excitation z , the intensity decreases until a new event occurs. Therefore $M = \mu + z$ is an upper bound on the intensity along the no-new-event path. Generate a candidate waiting time from an exponential clock of rate M .

$$\Delta \sim \text{Exp}(M), \quad z' = ze^{-\beta\Delta}, \quad \lambda' = \mu + z'. \quad (18.1)$$

Keep the candidate with probability λ' divided by M . Equivalently, draw a uniform U and retain it when UM is below λ' . A retained candidate becomes a transaction attempt and adds $\eta\beta$ to the residual. A discarded candidate advances time and decays the residual but produces no transaction.

$$\Pr(\text{retain} \mid \Delta) = \frac{\lambda'}{M}. \quad (18.2)$$

Why this works. A candidate occurs locally with probability $M dt$. Multiplying by its retention probability gives $\lambda' dt$, the desired event probability. Recomputing the bound after every candidate accounts for the evolving history. This local argument captures the mechanism; the formal result requires the bound to be predictable and to dominate the conditional intensity.

Concrete example. Suppose $\mu = 0.10$ and $z = 0.525$, so $M = 0.625$. A candidate arrives 0.40 hours later. The residual has fallen to about 0.28813, giving intensity 0.38813 and retention probability 0.6210. A uniform draw 0.50 retains it; a draw 0.80 discards it. If retained, the new residual becomes 0.81313.

Hour boundaries. If the next candidate lies beyond the end of the hour, no event is inserted there. The residual is decayed exactly to the boundary and passed to the next interval, where the background may change. Drawing a new candidate under the new bound is valid; carrying the old bound across a background jump would not generally be valid.

Meaning of rejection. Thinning rejection is a mathematical sampling operation. It is invisible in the transaction ledger. A declined payment, in contrast, is a real generated attempt that later fails the funds check.

Original simulation method: [Ogata \(1981\)](#).

17. A worked event-clock calculation

Consider one account at the start of an hour with background $\mu = 0.10$ and residual $z = 0.525$. Hold all latent states fixed during this hour. We follow two illustrative candidate draws to make the algorithm's time and memory updates explicit.

First candidate. With upper bound 0.625, choose an exponential draw of 0.40 hours and a uniform retention draw of 0.50. As on the previous page, this candidate is retained. It occurs at minute 24, and the residual immediately afterwards is approximately 0.81313.

Second candidate. The new bound is 0.91313. Suppose the next waiting-time draw is 0.20 hours. The residual decays to $0.81313 \exp(-0.30)$, approximately 0.60239. The retention probability is then about 0.7692.

$$\frac{0.10 + 0.81313e^{-1.5(0.20)}}{0.10 + 0.81313} \simeq 0.7692. \quad (19.1)$$

If the next uniform draw is 0.90, this second candidate is rejected. We are now 0.60 hours into the interval, with residual 0.60239 and no second transaction. Suppose the next exponential candidate would arrive 0.80 hours later, beyond the remaining 0.40 hours. The interval ends with residual about 0.33060.

$$z_{\text{end}} = 0.60239e^{-1.5(0.40)} \simeq 0.33060. \quad (19.2)$$

Behavioural interpretation. This hour contains one attempted payment, even though three candidate waiting times were drawn. The one attempt may eventually be legitimate or fraudulent, large or small, accepted or declined. None of those marks have yet been sampled in this clock calculation.

For each hour, the simulator generates each account's event times, merges them and orders them chronologically before applying marks and balance changes. This is valid because the within-hour event clocks do not depend on incoming balances or recipient choices. Those variables do affect the marks and settlement, so their updates must follow global event order.

The separation between a clock and its marks is useful experimentally: it lets us ask whether fraud is detectable from timing, payment attributes or network structure, while knowing exactly which mechanisms produced each kind of evidence.

18. Why recipient choice has a Gibbs form

At an event from sender i , the model chooses one other account j . Each candidate receives a dimensionless preference cost E . A lower cost means greater preference. The Gibbs distribution is a principled way to turn these relative costs into positive probabilities while retaining randomness.

$$p_j = \frac{e^{-E_j/T}}{Z}, \quad Z = \sum_{k \neq i} e^{-E_k/T}. \quad (20.1)$$

One derivation maximises Shannon entropy subject to normalisation and a fixed expected cost. Introduce multipliers α and γ and differentiate with respect to each probability.

$$\mathcal{L} = -\sum_j p_j \ln p_j - \alpha \left(\sum_j p_j - 1 \right) - \gamma \left(\sum_j p_j E_j - \bar{E} \right). \quad (20.2)$$

The stationarity equation is $-\ln p_j - 1 - \alpha - \gamma E_j = 0$. Exponentiating and normalising gives the Gibbs expression, with $\gamma = 1/T$. For the finite candidate set, entropy is concave, so this stationary solution is the entropy maximum under the specified constraints.

Concrete example. For three candidates with costs 0, 0.8 and 1.6 at $T = 0.8$, the unnormalised weights are 1, $\exp(-1)$ and $\exp(-2)$. Normalising gives approximately 0.6652, 0.2447 and 0.0900. The most preferred recipient is likely, but not certain.

Behavioural meaning. T controls how sharply choices follow the specified preferences. High T makes costs matter less; low T concentrates choices near the cheapest candidates. It is a dimensionless choice parameter, not a temperature in kelvin. The cost is not monetary expenditure or physical energy.

The maximum-entropy derivation explains a distributional form. The simulator specifies T and the costs directly; it does not measure a customer's expected cost and solve an inverse thermodynamic problem. Nor does local Gibbs choice establish a global equilibrium distribution for the evolving network.

Maximum-entropy foundation: [Jaynes \(1957\)](#). General concept: [MIT: information and entropy](#).

19. Community preference and probability ratios

Before fraud-specific preference is added, recipient cost combines community mismatch and previous sender-recipient interactions. The memory term uses the number of earlier attempts, not only settled payments.

$$E_{ij} = 1.4\mathbf{1}\{g_i \neq g_j\} - w\ln(1 + n_{ij}), \quad w = 1. \quad (21.1)$$

A useful simplification is to compare two recipients directly. Their partition function cancels. Adding the same constant to every cost also leaves probabilities unchanged; this permits numerically stable evaluation by subtracting the minimum cost.

$$\frac{p_j}{p_k} = \exp\left[-\frac{E_{ij} - E_{ik}}{T}\right]. \quad (21.2)$$

Concrete example. Compare an unfamiliar same-community recipient with an unfamiliar different-community recipient. Their costs are 0 and 1.4. At $T = 0.8$, the former is $\exp(1.75)$, about 5.75 times as likely as the latter. This is a pairwise ratio. The total probability of sending outside the community also depends on how many candidates are outside it.

For example, with two same-community candidates and six outside candidates, all unfamiliar and with no fraud preference, the total outside probability is $6 \exp(-1.75)$ divided by $[2 + 6 \exp(-1.75)]$, approximately 0.343. A preference against crossing communities does not make such transfers rare when many outside candidates are available.

Behavioural meaning. Community labels represent coarse affinity. They do not encode actual geography, family membership or occupational networks. Counts make relationships persistent: a familiar recipient can overcome a community mismatch if enough previous interactions reduce its cost.

At very high T the distribution approaches uniform over all other accounts. As T approaches zero from above, it concentrates on minimum-cost recipients, sharing weight among ties. The actual default $T = 0.8$ keeps exploration possible. Since costs depend on the accumulating history, a probability computed now is not a permanent property of an account pair.

The stable implementation uses $\exp[-(E - \text{minimum } E)/T]$. The common factor cancels exactly under normalisation; this changes numerical conditioning, not the model.

20. Reinforcement creates network memory

Substituting the cost into the Gibbs weights reveals the memory mechanism directly. Ignoring the fraud term for a moment, a recipient's unnormalised weight factorises into community affinity and a power of previous interaction count.

$$q_{ij} = e^{-1.4\mathbf{1}\{g_i \neq g_j\}/T} (1 + n_{ij})^{w/T}. \quad (22.1)$$

At the default parameters, $w/T = 1.25$. The additive one ensures that a never-used edge still has positive weight. Without it, a positive power of zero would prevent any new recipient from ever being selected.

Concrete example. Compare two same-community recipients with prior counts nine and zero. Their weights are 10 to the power 1.25 and one, approximately 17.78 and one. If they were the only candidates, the familiar recipient would have probability $17.78/18.78 = 0.9468$. With many other accounts, its absolute probability falls, but this pairwise ratio remains unchanged.

One additional attempt changes an existing edge's memory weight by the factor below.

$$\frac{q(n+1)}{q(n)} = \left(\frac{n+2}{n+1}\right)^{1.25}. \quad (22.2)$$

The first attempt changes weight from one to about 2.378. Changing count nine to ten multiplies it by about 1.1265. Early interactions can therefore strongly influence subsequent choices, even though the marginal multiplicative gain decreases with count.

Behavioural meaning. Customers develop familiar counterparties. Chance early choices can become persistent habits, making different seeded worlds diverge in network structure. This is path dependence: two accounts with identical fixed parameters can accumulate different interaction patterns.

Memory does not decay and counts include declined attempts. Consequently the model describes familiarity with attempted counterparties, not solely successful financial relationships. It has no explicit forgetting, relationship deletion or saturation. The power-law form of the weight does not prove that the resulting graph has a power-law degree distribution; that would require a separate analysis.

Setting $w = 0$ removes reinforcement while preserving community preference and fraud-specific routing. This is the network-memory ablation used to test how much the experiment depends on historical recipient choice.

21. The hidden recipient ring

A subset R of accounts is selected uniformly without replacement at initialisation. Its size is the larger of three and the integer part of $N/20$. A 64-account run therefore has three selected accounts; a 160-account run has eight.

For fraudulent attempts only, the cost of sending to a member of R is reduced. The effective bonus decreases after the environmental shift.

$$E_{ij}^{(Y)} = 1.4\mathbf{1}\{g_i \neq g_j\} - w\ln(1 + n_{ij}) - Y \frac{3.5}{1+d} \mathbf{1}\{j \in R\}. \quad (23.1)$$

Concrete example. Before the shift, two otherwise identical recipients have a ring-to-non-ring probability ratio $\exp(3.5/0.8)$, approximately 79.44, for a fraudulent attempt. After d becomes one, the ratio falls to $\exp(1.75/0.8)$, about 8.91. The preference remains but becomes much less concentrated.

As a deliberately simplified illustration, take a sender outside the ring and suppose all 63 candidates have identical community and memory costs. Three candidates are in R . Before the shift, the total ring probability is approximately 3 times 79.44 divided by $[3 \text{ times } 79.44 + 60]$, or 0.799. Afterwards it is approximately 0.308. Actual probabilities vary with communities, memory and whether the sender itself is in R .

Behavioural meaning. Fraudulent traffic is preferentially concentrated on a hidden recipient subset, creating a recoverable network signature. Membership alone does not make all incoming or outgoing transactions fraudulent. The sender's compromise state and the fraud draw determine Y first.

Despite the name ring, R is not a preconstructed cycle. The model does not force A to pay B , B to pay C and C to pay A . Nor does membership increase compromise hazard or transmit compromise along edges. Any observed cycle emerges from realised transactions and recipient preferences.

The descriptive typology marks a fraudulent event from a sender in R as a mule transfer. This label is a project convention, not proof that specific funds are being laundered or forwarded. Amount provenance and coordinated criminal plans are not tracked.

22. Amounts, tails and fraud multipliers

Once the fraud label and receiver have been sampled, the attempted amount is drawn. Define f as the fraud multiplier, equal to one for legitimate events and $2.8/(1+1.4d)$ for fraudulent events. Before rounding, the log-amount distribution is Gaussian.

$$\ln(A/L) \mid x, Y, d \sim \mathcal{N}(x + \ln f, 0.60^2). \quad (24.1)$$

Every conditional quantile is multiplied by f . Writing Φ for the standard Gaussian distribution function gives a direct expression for the probability of an unusually large attempt.

$$\Pr(A > a \mid x, Y, d) = 1 - \Phi\left(\frac{\ln(a/L) - x - \ln f}{0.60}\right). \quad (24.2)$$

Concrete example. At $x = \ln 45$ before the shift, ordinary amounts have median 45 and fraudulent amounts median 126. Their means are 53.87 and 150.85. For a 100-unit threshold, the ordinary upper-tail probability is approximately 9.2%, while the fraudulent probability is approximately 65.0%. A large amount is informative but is not a perfect fraud rule.

At $d = 1$, the fraud multiplier becomes $2.8/2.4 = 1.1667$. Holding x fixed, the fraudulent median falls to 52.50. The simultaneous target shift can subsequently raise x , but its effect is gradual and separate. Combining these effects without specifying time since the shift would give a misleading single number.

Recorded amounts use integer minor units: multiply the continuous major-unit amount by 100, round to the nearest integer and enforce a minimum of one. The analytical lognormal equations describe the continuous precursor. Rounding creates a discrete distribution and slightly changes exact moments, particularly for extremely small values.

Behavioural meaning. Fraud is generated as an overlapping distribution, not a separate range of impossible legitimate amounts. Legitimate high-spending episodes can resemble fraud; compromised accounts can produce small fraudulent attempts. This overlap is central to why detection is a statistical inference problem.

Tail and transformation conventions: NIST: lognormal distributions. The multiplier values and their shift are simulator assumptions.

23. Foreign and new-device indicators

Two binary marks create additional observable evidence. Their probabilities depend on the fraud label and, for foreign activity, the behavioural state. They are indicators only: the simulator does not maintain a country itinerary or an inventory of physical devices.

For a non-travelling account, foreign probability is $0.55/(1+d)$ for fraud and $0.04 + 0.03 \min(d,2)$ for legitimate activity. For a travelling account, it is 0.45 regardless of Y . New-device probability is $0.65/(1+d)$ for fraud and 0.035 otherwise.

$$\Pr(D = 1 \mid Y = 1, d) = \frac{0.65}{1+d}, \quad \Pr(D = 1 \mid Y = 0) = 0.035. \quad (25.1)$$

Concrete example. Before the shift, a non-travelling fraudulent attempt is foreign with probability 0.55 and uses a new device with probability 0.65. Conditional on this state and label, the draws are independent, so both occur with probability 0.3575. For a non-travelling legitimate attempt, the corresponding probability is 0.04 times 0.035 = 0.0014.

Conditional independence does not imply independence after pooling over hidden labels and states. Let p be fraud prevalence in an illustrative fixed non-travel mixture. The covariance induced solely by label mixing is:

$$\text{Cov}(F, D) = p(1 - p)(f_1 - f_0)(d_1 - d_0). \quad (25.2)$$

With $p = 0.02$ and pre-shift probabilities, this is about 0.00615. The value 2% here is an illustrative mixture assumption, not the simulator's analytically guaranteed fraud prevalence.

Behavioural meaning. Correlated warning signs can arise because they share a hidden cause. A detector need not see compromise itself to exploit these consequences. Travel supplies a legitimate explanation for foreign activity, preventing a simplistic foreign-equals-fraud interpretation.

After $d = 1$, fraudulent new-device probability falls to 0.325 and non-travel fraudulent foreign probability to 0.275, while non-travel legitimate foreign probability rises to 0.07. The model deliberately makes some formerly useful distinctions weaker. Because the travel branch is evaluated first, travelling fraudulent attempts retain foreign probability 0.45.

24. The ledger as a conservation law

An attempted payment is settled only if the sender has enough funds immediately before the event. The receiver does not decide whether to decline it, and the fraud detector does not block it. Let Q denote settlement and A the amount in integer minor units.

$$Q = \mathbf{1}\{B_i(t^-) \geq A\}, \quad (26.1)$$

$$B(t^+) = B(t^-) + QA(e_j - e_i). \quad (26.2)$$

Here the basis vector e with subscript j selects the recipient account. Summing the update over accounts gives zero, proving conservation of total money for every event. The funds condition also preserves nonnegative balances if initial balances are nonnegative.

$$\sum_i B_i(t) = \sum_i B_i(0), \quad B_i(t) \geq 0. \quad (26.3)$$

Concrete example. If A has balance 5,000 minor units and attempts 6,000, the payment is declined and neither balance changes. The attempt remains in the event history and increments sender-recipient memory. If A instead attempts 4,000 to B, A's balance falls by 4,000 and B's rises by 4,000. A subsequent event sees these updated balances.

Behavioural meaning. The ledger imposes a hard capacity constraint on successful movement of funds while leaving attempted behaviour unconstrained. Repeated large attempts can deplete a sender's balance and produce later declines. Incoming transfers can replenish it. This creates history-dependent settlement even though the amount distribution itself is not truncated to available funds.

Conditioning analysis on settled payments introduces selection: successful amounts are filtered by the current balance. A low observed settled amount need not imply a low attempted spending scale. Similarly, prevented-by-insufficient-funds fraud differs from model-detected fraud.

The system is closed. It has no salary injection, withdrawal to the external world, overdraft, fees or exchange-rate conversion. Integer arithmetic avoids floating-point rounding drift in the conservation law. Conservation is a useful software invariant; it does not make the system a realistic complete banking economy.

25. Entropy measures recipient diversity

For one sender, let n with subscripts ij count previous attempts to recipient j and let n with subscript i be their total. The empirical recipient probabilities are counts divided by that total. Shannon entropy measures how spread out the observed choices are.

$$\hat{p}_{ij} = \frac{n_{ij}}{n_i}, \quad H_i = - \sum_{j: n_{ij} > 0} \hat{p}_{ij} \ln \hat{p}_{ij}. \quad (27.1)$$

Using natural logarithms gives entropy in nats. Zero-count terms contribute zero by continuity. With no previous attempts, the implementation uses zero as a practical initial value rather than trying to estimate a distribution from no observations.

Concrete example. Four prior attempts all to B have entropy zero. Two to B and two to C have entropy $\ln 2 = 0.6931$. One each to four different recipients has entropy $\ln 4 = 1.3863$. Counts nine and one give entropy about 0.3251: two recipients have been used, but the relationship is strongly concentrated.

An equivalent expression avoids repeatedly normalising all counts and makes an efficient cumulative calculation possible.

$$H_i = \ln n_i - \frac{1}{n_i} \sum_j n_{ij} \ln n_{ij} \quad (n_i > 0). \quad (27.2)$$

Behavioural meaning. Low entropy indicates concentrated historical recipient choice; high entropy indicates broader use. It does not identify criminal intent. Paying rent repeatedly can create low entropy, and a fraudulent account can distribute payments across many recipients. Entropy is evidence to combine with other features.

The empirical entropy here differs from the entropy of the current Gibbs choice distribution. The former summarises realised history; the latter describes the probabilities of the next draw. A change in current preferences may take time to become visible in the expanding historical estimate.

All past attempts contribute and there is no forgetting window for this feature. Consequently an abrupt behavioural change may be diluted by a long previous history. This is a concrete mechanism by which a model's observable features can respond more slowly than the latent environment itself.

Entropy originates in [Shannon \(1948\)](#). General conceptual material: [MIT: information and entropy](#).

26. Temporal loops are ordered events

A static graph records that account pairs interacted, but discards when those interactions happened. A temporal loop requires a sequence of settled transfers whose directions connect and whose timestamps increase. The replay highlights selected loops of length two or three within a 24-hour window.

$$A \xrightarrow{t_1} B \xrightarrow{t_2} C \xrightarrow{t_3} A, \quad t_1 < t_2 < t_3, \quad t_3 - t_1 \leq 24. \quad (28.1)$$

Concrete example. A pays B at hour 10, B pays C at hour 12, and C pays A at hour 15. The final event closes a three-edge temporal loop. If B paid C at hour 9 instead, the aggregated graph would still contain a triangle, but this particular chronological sequence would not form the same time-respecting loop.

A reverse payment can close a two-edge loop: A pays B at hour 10 and B pays A at hour 11. The replay gives a two-edge loop preference when both a reverse edge and a three-edge candidate are available. It reports one representative loop per closing event, not every possible cycle in the graph.

Behavioural meaning. Loops display recirculation of transfers and can suggest patterns worth investigating. They are not proof of laundering. The amounts need not match, and the ledger does not tag individual units of money with provenance. A owes B for dinner and B later reimburses A for tickets: the same topological pattern can be legitimate.

Declined attempts are excluded from replay loops because no funds moved. They remain included in other histories, such as recipient familiarity and attempt counts. Thus the settlement network used for loop visualisation is different from the attempted-interaction network used by several detector features.

Loop highlighting is explanatory visualisation; it is not currently an input feature to the classifiers. More elaborate motifs, longer cycles, exhaustive enumeration and amount-conserving flow tracing would require additional definitions and algorithms.

Why time ordering matters for networks: [Holme & Saramaki \(2012\)](#). This replay's 24-hour rule and short-loop restrictions are project choices.

27. The joint generative structure

The simulation is a hierarchy of conditional distributions. First draw fixed customer attributes. Then update hourly latent states. Conditional on these states and each sender's excitation history, generate event times. Finally, draw marks and deterministically update balances and interaction counts.

For an event from sender i at t , let the conditioning information include the current latent state, time, sender and preceding history. The joint mark distribution is the product of the two conditional factors below, with this information denoted by a calligraphic $\mathcal{L}$.

$$p(Y, j, A \mid \mathcal{L}) = p(Y \mid C_i) p(j \mid Y, \mathcal{L}) p(A \mid Y, x_i, d) \quad (29.1)$$

$$p(F, D \mid Y, \mathcal{L}) = p(F \mid Y, S_i, d) p(D \mid Y, d). \quad (29.2)$$

The ledger then determines settlement Q from amount and pre-event balance. It is not another independent Bernoulli draw. This ordering explains why the simulator knows the true fraud label: it is a generative cause of some marks, rather than a judgement inferred afterwards from them.

Concrete example. Suppose the sender is compromised. First the fraud draw may produce $Y = 1$. That single hidden outcome then strengthens ring preference, increases the amount multiplier and changes device and foreign probabilities. These observables become associated through a shared cause, even where their random draws are conditionally independent.

What is Markovian? The full system can be expressed using an augmented state containing current regimes, log scales, residual excitation, balances and pair counts. Given that state and the time-dependent environment, older details need not be retained to generate the next update. Observing only raw amounts does not reveal this augmented state, so the amount series alone generally is not a first-order Markov process.

Behavioural meaning. Controlled simulation lets us know which mechanism caused a statistical association. It also lets us remove one mechanism and observe what changes. This is stronger than inspecting correlations in an unexplained synthetic table, but weaker than establishing a causal mechanism in a real bank.

The three random streams for latent states, event times and marks improve reproducibility and separation of concerns. They do not make all observed variables independent: the equations deliberately transmit dependence from latent states into events and marks.

28. One transaction, from state to settlement

Consider a worked event at 15:00 before the shift. A customer has baseline scale 45, activity multiplier 2, prior ordinary behaviour and no residual excitation. Suppose its new hourly state is travel and it is compromised. These are assumed realised draws for illustration, not the most probable path.

Spending update. The new target is $\ln 45 + 0.25$. Start x at $\ln 45$ and choose OU innovation $Z = 0.5$. The updated log-scale increase is:

$$0.25(1 - e^{-0.06}) + 0.116488(0.5) \simeq 0.072803. \quad (30.1)$$

The resulting ordinary spending scale is about 48.40. The background event rate is 0.10 times 2 times 1.55 times 1.4 times 3 = 1.302 per hour. Suppose the stochastic clock generates an event during this hour; x and the background factors remain fixed until the next hourly update.

Fraud and recipient. A fraud uniform draw of 0.40 is below 0.65, so $Y = 1$. The receiver is sampled from the entire Gibbs distribution, including community, prior counts and ring preference. Suppose the chosen receiver is B. Being in the ring makes selection more likely under fraud; it does not force this particular outcome.

Amount and binary marks. Choose event-level log-noise $\epsilon = 0.30$. The pre-rounding amount is approximately 48.40 times $\exp(0.30)$ times 2.8, or 182.93 currency units. A foreign uniform draw 0.40 falls below the travel probability 0.45, and a device draw 0.20 falls below 0.65, so both flags are one.

Settlement and history. If the sender has 20,000 minor units, the roughly 18,293-unit attempt settles, leaving roughly 1,707. The recipient gains exactly the debited integer amount. Pair count rises by one, and the attempt has already added 0.525 per hour of excitation to the sender's clock. Its label becomes available for training 24 hours after its timestamp.

A different noise draw could make this event small; a smaller starting balance could make it decline. The same hidden compromise episode therefore produces a distribution of observable outcomes rather than one fixed fraud signature.

29. Distribution shift as a controlled intervention

The environment changes at a chosen fraction of the run. With default drift fraction 0.75, define d to be zero before the change and the configured strength afterwards. Because updates occur on integer hours, a non-integer threshold takes effect at the first integer hour at or above it.

$$d(h) = d_0 \mathbf{1}\{h \geq 0.75H\}, \quad d_0 = 1. \quad (31.1)$$

For a 30-day run, $H = 720$ hours and the change is at hour 540, or 22.5 elapsed days. With Day 1 starting at hour zero, this is Day 23 at noon. The simulation clock and the page's display label should therefore not be confused.

Several mechanisms change together. Compromise hazard doubles. The compromised activity multiplier falls from 3 to 2. The log-spending target increases by 0.20 for all accounts. The fraudulent amount multiplier falls from 2.8 to 1.1667. The ring bonus falls from 3.5 to 1.75. Device and non-travel foreign indicators also become less separated by label.

Concrete example. Immediately after the shift, two otherwise identical compromised accounts have a lower background multiplier than they would have had under $d = 0$. At the same time, more safe accounts become compromised. It is therefore incorrect to infer total fraudulent transaction volume from the activity multiplier alone. The occupancy distribution also needs time to adapt.

Behavioural meaning. The post-shift environment contains more exposure to compromise but some subtler fraud signatures. The resulting challenge tests whether a detector trained earlier continues to rank and flag events usefully when the generative law changes.

This is a bundled intervention, not a clean isolation of one textbook shift category. Covariate shift changes observable inputs while preserving label probabilities conditional on them; prior shift changes class proportions under additional assumptions; concept shift changes the conditional label relation. Here several latent and mark mechanisms change simultaneously, so those invariances should not be assumed.

Nor is this an adaptive adversary responding to the fitted classifier. The shift is scheduled independently of model performance. That makes experiments controlled and reproducible, while limiting claims about strategic fraud adaptation.

30. Timescales and dimensionless comparisons

Physics offers a useful habit: compare timescales before interpreting a coupled system. A parameter matters partly through its ratio to the time over which observations are collected. For the current defaults, excitation fades much faster than spending tendencies relax.

$$\tau_{\text{kernel}} = \beta^{-1} = 0.667 \text{ h}, \quad \tau_x = \kappa^{-1} = 16.667 \text{ h}. \quad (32.1)$$

The compromised episode mean is about 10.51 hours; travel episodes average 25 hours; ordinary episodes average 100 hours. The daily forcing period is 24 hours and labels arrive after 24 hours. These different scales explain why a one-hour feature, a one-day feature and an expanding-history feature can reveal different aspects of the same event.

Dimensionless ratios distinguish regimes. For example, beta divided by kappa is 25, indicating fast excitation relative to log-spending relaxation. The product kappa times 24 is 1.44, so the log-spending memory of a fixed-target displacement falls to $\exp(-1.44)$, about 0.237, across a day.

$$\frac{\beta}{\kappa} = 25, \quad \kappa(24 \text{ h}) = 1.44, \quad \frac{w}{T} = 1.25. \quad (32.2)$$

Concrete example. Increasing beta from 1.5 to 3 while keeping eta fixed doubles the instantaneous excitation jump but halves its decay time. The total kernel area remains 0.35. Thus transactions cluster more tightly without changing the constant-background stationary mean rate. A detector using a one-hour count can respond differently even when total daily volume stays similar in that reference model.

Behavioural meaning. Timing resolution should match the phenomenon. Daily aggregation can hide a 20-minute burst. A long expanding recipient history can hide a recent change. Conversely, a very short window may contain too few events for a stable estimate.

The model also contains a numerical structural scale: latent states are updated hourly. Within-hour interpolation of the OU path or a finer behavioural chain would define a different process unless its parameters and update rules were adjusted consistently. Increasing animation speed changes none of these scientific timescales; it only changes how quickly a stored history is displayed.

31. Monte Carlo means more than one run

A seed selects one realisation of the random world. To estimate a property of the generator, repeat the experiment over independent seeds and examine variation across runs. A realised fraud rate, loop count or detector score is a statistic of one world, not a universal constant of the parameter configuration.

For independent run-level values M with subscripts r , use the sample mean and sample variance. Their standard error quantifies uncertainty in the estimated mean across seeded worlds.

$$\bar{M} = \frac{1}{R} \sum_{r=1}^R M_r, \quad s_M^2 = \frac{1}{R-1} \sum_{r=1}^R (M_r - \bar{M})^2, \quad (33.1)$$

$$\text{SE}(\bar{M}) = \frac{s_M}{\sqrt{R}}. \quad (33.2)$$

Concrete example. Suppose five runs give recall values 0.40, 0.50, 0.45, 0.55 and 0.60. Their mean is 0.50, sample standard deviation about 0.0791 and standard error about 0.0354. An approximate 95% Student-t interval uses multiplier 2.776 for four degrees of freedom, giving roughly 0.402 to 0.598. This is an illustrative calculation, not a reported project result.

Behavioural meaning. Some synthetic worlds are harder because they contain different compromised accounts, early network choices or numbers of positive test events. A detector that wins on one seed may not win on average. With only five runs, uncertainty can be substantial, and normality-based intervals deserve caution for bounded or highly skewed metrics.

Transactions within a run are dependent: Hawkes bursts, repeated senders and network reinforcement connect observations. Treating thousands of transactions as thousands of independent worlds would underestimate uncertainty. Run-level replication respects the simulation's experimental unit.

A fixed seed supports reproducibility in a fixed software environment and execution path. It is not evidence of robustness across environments or model assumptions. Good reporting includes parameters, seed set, software versions and the distribution of outcomes, rather than only the most favourable run.

General Monte Carlo principles and estimators: [Owen: Monte Carlo theory, methods and examples](#).

32. Ablations, sensitivity and identifiability

An ablation removes one mechanism and compares the resulting experiment. Current examples set η to zero to remove self-excitation, w to zero to remove recipient memory, or d to zero to remove the environmental shift. These interventions answer different scientific questions.

Concrete example. With constant baseline 0.10, removing $\eta = 0.35$ reduces the stationary reference rate from 0.15385 to 0.10 per hour. If detection performance changes, the cause may include both loss of burst structure and fewer training examples. A volume-matched extension could set the no-excitation baseline to 0.15385, but that is a distinct experiment and is not the existing ablation.

When two detectors are evaluated on the same seeded world, their performance difference is naturally paired. Its uncertainty should be calculated from per-world differences, preserving shared difficulty.

$$\Delta_r = M_{A,r} - M_{B,r}, \quad \text{SE}(\bar{\Delta}) = \frac{s_{\Delta}}{\sqrt{R}}. \quad (34.1)$$

Using the same seed for two different simulator configurations does not guarantee an event-by-event counterfactual. A changed event count consumes random draws differently and alters later histories. Separate random streams preserve some structure, especially latent dynamics when only event or mark mechanisms change, but do not make all events correspond.

Identifiability. A mean event rate alone identifies only the ratio $\mu/(1-\eta)$ in the constant-background reference. Many parameter pairs yield that mean. Likewise, OU stationary variance identifies σ^2 divided by 2κ , not both parameters independently. Temporal correlations, cluster structure and other summaries are needed to distinguish them.

Behavioural meaning. Matching one histogram is a weak calibration target. A realistic model would need to match several temporal, cross-sectional and network statistics simultaneously, then be tested on summaries not used in fitting. Otherwise different underlying mechanisms can look similar in the selected view.

These considerations separate verification from validation. Verification asks whether the program samples the equations correctly. Validation asks whether those equations are useful representations of the intended financial setting. The latter requires external evidence that this synthetic-only project does not supply.

33. Mathematical checks on a generated world

Known analytical properties provide tests of implementation. The ledger's total money must remain exactly constant. Every account balance must stay nonnegative. Every probability vector must sum to one. Event timestamps must be chronological, and feature histories must exclude the event currently being scored.

Stochastic checks differ from exact invariants. An empirical OU mean should approach its analytical target within sampling uncertainty; it should not equal it exactly. Similarly, event counts from a Hawkes reference simulation should fluctuate around their expected value. A test that demands exact equality of random moments is mathematically inappropriate.

Concrete example. For independent one-hour OU transitions beginning exactly at a fixed target, the innovation mean is zero and variance about 0.0135695. With 10,000 independent draws, the standard error of their mean is approximately $0.116488/100 = 0.001165$. A sample mean of 0.001 is unremarkable; a mean near 0.10 suggests a serious problem.

A more advanced diagnostic uses the integrated conditional intensity. For a correctly specified point process under suitable regularity conditions, transformed inter-event intervals follow unit-rate exponential behaviour.

$$u_k = \int_{t_{k-1}}^{t_k} \lambda(s) ds. \quad (35.1)$$

For one sender, the event-time log-likelihood has an event reward and an integrated-rate penalty. With known latent backgrounds it can support further checks or inference.

$$\ell = \sum_k \ln \lambda(t_k) - \int_0^H \lambda(s) ds. \quad (35.2)$$

Scope. The integrated-intensity diagnostic and likelihood are mathematical extensions explained here; the current evaluation pipeline does not fit Hawkes parameters or run this diagnostic. They should not be reported as completed validation results.

Behavioural meaning. Correct random generation is more than producing plausible-looking animation. Numerical checks establish that the intended law is sampled, while visualisation helps explain individual paths. Neither substitutes for empirical validation against real-world data.

Conditional-intensity simulation and likelihood context: [Ogata \(1981\)](#); broader point-process treatment: [Bacry, Mastromatteo & Muzy \(2015\)](#).

34. What a detector is allowed to observe

The generator sees hidden states, but the detector receives only observable event attributes and causal summaries of earlier attempts. This separation makes detection a meaningful inference task. Giving the detector ring membership or compromise state would disclose the mechanism it is supposed to infer.

The current feature groups are transaction attributes, temporal behaviour and recipient-network history. Transaction attributes include log amount, time of day, foreign and device indicators, and amount relative to the sender's pre-event balance. Temporal summaries include previous one-hour and 24-hour counts, time since the last attempt and a historical amount deviation.

For at least two previous sender observations, the deviation score uses the previous sample mean and sample standard deviation of transformed amounts. A floor of 0.3 prevents division by very small historical variation.

$$z_A = \frac{\ln(1 + A/L) - \bar{\ell}_{\text{past}}}{\max(s_{\text{past}}, 0.3)}. \quad (36.1)$$

Concrete example. If the previous transformed-amount mean is 3.8 and standard deviation 0.2, a current transformed amount of 4.4 gives deviation $(4.4 - 3.8)/0.3 = 2.0$. The floor makes this less extreme than dividing by 0.2. With fewer than two historical observations, the project uses zero for this feature.

Network features include whether the pair is new, its prior attempt count, the receiver's number of distinct prior senders and the sender's expanding recipient entropy. They are all computed before inserting the current event into history.

Behavioural meaning. The same amount can be routine for one account and unusual for another. A new recipient means something different after hundreds of stable payments than it does for a newly observed customer. Features convert such contextual distinctions into quantities a statistical model can use.

The feature logarithm $\ln(1 + A/L)$ is not identical to the generator's $\ln(A/L)$; the added one is a numerical transformation for observable amounts. Neither raw account identifiers, true fraud labels, hidden regimes, ring membership nor future settlement outcomes are supplied as detector inputs.

35. Delayed labels and chronological training

The simulator knows Y immediately because it generated it. The experimental learner is allowed to use that label only after a delay δ , defaulting to 24 hours. This creates two distinct information clocks: event arrival and label availability.

$$t_{\text{label}} = t_{\text{event}} + \delta. \quad (37.1)$$

Let training end at $0.50H$ and validation end at $0.70H$. Training requires both an event before the training boundary and a label available by that boundary. Validation events belong to the next interval and also require maturity by its endpoint. Test events begin at $0.70H$.

$$\mathcal{T}_{\text{train}} = \{k : t_k < 0.50H, t_k + \delta \leq 0.50H\}. \quad (37.2)$$

Concrete example. For $H = 720$ hours, training ends at hour 360. With a 24-hour delay, events later than hour 336 cannot contribute labelled training examples at that time. Validation starts at 360 and ends at 504, so its eligible event times extend only to hour 480. Testing starts at hour 504, while the default shift arrives later at hour 540.

History is not erased during these label-maturity gaps. An unlabelled event can legitimately contribute to later counts, balances and recipient histories because its transaction attributes have already been observed. Its unknown label cannot be used as a predictor or training target before maturity.

Behavioural meaning. Realistic evaluation respects what would have been knowable when a decision was made. Randomly shuffling a dependent temporal dataset can mix future regimes and histories into training, making performance look more stable than it is.

The fitted models are frozen before replay. Seeing Day 9 in the animation does not mean a model is currently training on Day 9. The application is displaying a previously generated and evaluated timeline. Ground-truth colours can be shown immediately for explanation even though the training experiment imposes delayed labels.

Delayed supervision is an established practical concern in fraud evaluation: [Dal Pozzolo et al. \(2015\)](#). This project's fixed-delay, frozen-model protocol is simpler than the feedback systems studied there.

36. The role of machine learning

Supervised models use mature labelled examples to learn a relationship between observable features and Y . Logistic regression fits a linear score in transformed feature space; its logistic link turns that score into a probability. Gradient boosting combines nonlinear tree-based updates. Both learn from labels in the training period.

$$\hat{p}(Y = 1 | f) = \frac{1}{1 + e^{-(b + w^T f)}}. \quad (38.1)$$

The experiment compares full-feature boosting with versions excluding network information or using transaction attributes alone. These feature ablations ask which observable consequences of the simulator's mechanisms are useful. They do not remove those mechanisms from the generator itself.

Isolation Forest instead learns how readily observations can be isolated by random feature splits. Short average isolation paths suggest unusual points. It does not need fraud labels for fitting and is trained here on the mixed training feature population, not a guaranteed-clean set. A behavioural rule uses an amount-deviation score plus device and foreign indicators without learning parameters.

Concrete example. Suppose 100 test events contain two true frauds. A model that flags nothing has 98% accuracy but zero recall. A model flagging four events, including both frauds, has precision $2/4 = 50\%$ and recall $2/2 = 100\%$. The usefulness of a detector therefore cannot be judged by accuracy alone.

$$\text{precision} = \frac{TP}{TP + FP}, \quad \text{recall} = \frac{TP}{TP + FN}. \quad (38.2)$$

Behavioural meaning. Models are experimental probes of the environment. A performance drop after the shift may reveal dependence on old amount, device or network signatures. A strong result on one constructed world demonstrates recovery of its statistical structure, not deployment readiness at a bank.

The project reports average precision, ROC AUC and budget-based metrics alongside thresholded measures. A fixed threshold and retrospective daily top-K ranking answer different operational questions; neither changes the simulated ledger. Empty-positive segments require care because some metrics are undefined.

Why precision-recall evaluation is useful under class imbalance: [Saito & Rehmsmeier \(2015\)](#). Detailed estimator configuration belongs to the implementation companion.

37. What the mathematics does and does not establish

The model's strengths are explicit assumptions, known ground truth and separable mechanisms. It contains genuine stochastic processes and meaningful physics connections: restoring drift versus noise, excitation and branching, entropy-based probabilistic choice, multiple timescales and a conservation law. These connections do useful work in the generator.

Concrete example. Increasing temperature makes recipient choice less concentrated for a fixed cost vector. That conclusion follows from the local choice law. It does not imply that the whole transaction network approaches a thermodynamic equilibrium, because costs change as history accumulates and events are driven by nonstationary latent states.

Likewise, increasing a spending scale raises conditional amount quantiles but does not increase balance. Increasing η amplifies a constant-background clock but does not specify a real fraud mechanism. Highlighting a temporal loop reveals an ordered transfer pattern but does not trace the same money around that loop.

Main limitations. Account creation, closure, salaries, external merchants, exchange rates, geographic routes and strategic adaptation are absent. Community and ring membership are synthetic. Recipient memory never decays. The fixed-delay label model has no analyst errors or selective investigation. The closed ledger can produce balance depletion patterns unlike an open financial system.

How to describe the project accurately. It is a controlled stochastic laboratory for investigating how temporal and network structure affect fraud detection under a scheduled distribution change. The chosen parameters define scenarios. They have not been calibrated to a representative financial dataset, and the simulator does not establish real-world detection accuracy.

A useful next scientific step would be to select a specific observable target, such as inter-event correlations or repeated-recipient frequencies, obtain an appropriate empirical benchmark, and assess which assumptions reproduce it. That is a validation programme, not a cosmetic increase in model complexity.

The mathematical understanding to retain is the conditional sequence: customer heterogeneity creates different baselines; latent dynamics change local tendencies; event clocks create irregular attempts; probabilistic marks create observable evidence; the ledger imposes capacity constraints; and frozen detectors try to infer hidden fraud from the resulting history.

References I: stochastic processes

- [1] Uhlenbeck, G. E. and Ornstein, L. S. (1930). On the Theory of the Brownian Motion. *Physical Review*, 36, 823-841. DOI: [10.1103/PhysRev.36.823](https://doi.org/10.1103/PhysRev.36.823). [University-hosted full text](#). Historical foundation for the mean-reverting Gaussian process; see pages 10-13 for the project's derivation and restrictions.
- [2] Aldous, D. and Fill, J. A. *Reversible Markov Chains and Random Walks on Graphs*, online monograph, Chapter 2, section 2.1. [Aldous & Fill: finite Markov chains](#). Used for finite-chain notation and stationary-distribution concepts, not for the project's chosen transition probabilities.
- [3] Hawkes, A. G. and Oakes, D. (1974). A cluster process representation of a self-exciting process. *Journal of Applied Probability*, 11(3), 493-503. DOI: [10.2307/3212693](https://doi.org/10.2307/3212693). Foundational citation for the branching interpretation. The guide derives its elementary mean-cluster calculation explicitly.
- [4] Ogata, Y. (1981). On Lewis' simulation method for point processes. *IEEE Transactions on Information Theory*, 27(1), 23-31. DOI: [10.1109/TIT.1981.1056305](https://doi.org/10.1109/TIT.1981.1056305). [University-hosted full text](#). The PDF filename differs from the publication year; the article itself is dated 1981. Relevant to thinning and conditional-intensity reasoning.
- [5] Bacry, E., Mastromatteo, I. and Muzy, J.-F. (2015). Hawkes processes in finance. *Market Microstructure and Liquidity*, 1(1), 1550005. [Author preprint](#); [full text in HTML](#). A review connecting the process to financial event modelling; market-microstructure applications do not validate this simulator's retail assumptions.
- [6] Owen, A. B. (2013). *Monte Carlo theory, methods and examples*. Online book. [Owen: Monte Carlo theory, methods and examples](#). General reference for simulation estimators and uncertainty. The five-run numerical example is constructed for this guide.
- General concept links.** [NIST: lognormal distributions](#); [NIST: exponential distributions](#); [MIT: Brownian motion and mean reversion](#). These provide distribution conventions and accessible background. All lognormal quantities in the guide specify which location, standard deviation and conditioning level are intended.

References II: networks and inference

- [7] **Jaynes, E. T. (1957).** Information Theory and Statistical Mechanics. *Physical Review*, 106, 620-630. DOI: [10.1103/PhysRev.106.620](https://doi.org/10.1103/PhysRev.106.620). [University-hosted full text](#). Foundation for maximum-entropy reasoning; the recipient cost is this project's own specification.
- [8] **Shannon, C. E. (1948).** A Mathematical Theory of Communication. *Bell System Technical Journal*, 27, 379-423 and 623-656. [University-hosted full text](#). Entropy as a measure of distributional spread. General learning resource: [MIT: information and entropy](#).
- [9] **Holme, P. and Saramaki, J. (2012).** Temporal networks. *Physics Reports*, 519(3), 97-125. DOI: [10.1016/j.physrep.2012.03.001](https://doi.org/10.1016/j.physrep.2012.03.001). [Author preprint](#). Background on time-respecting network structure. The replay's selected two- and three-edge loops are a narrower project-specific construction.
- [10] **Perra, N., Goncalves, B., Pastor-Satorras, R. and Vespignani, A. (2012).** Activity driven modeling of time varying networks. *Scientific Reports*, 2, 469. DOI: [10.1038/srep00469](https://doi.org/10.1038/srep00469). [Perra et al. \(2012\)](#). Related perspective on heterogeneous agent activity. FluxFraud additionally uses sender excitation and accumulated recipient reinforcement.
- [11] **Dal Pozzolo, A., Boracchi, G., Caelen, O., Alippi, C. and Bontempi, G. (2015).** Credit card fraud detection and concept-drift adaptation with delayed supervised information. *International Joint Conference on Neural Networks*. DOI: [10.1109/IJCNN.2015.7280527](https://doi.org/10.1109/IJCNN.2015.7280527). [Dal Pozzolo et al. \(2015\)](#). Motivation for respecting delayed supervision; the project does not reproduce the paper's full adaptation system.
- [12] **Saito, T. and Rehmsmeier, M. (2015).** The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, 10(3), e0118432. [Saito & Rehmsmeier \(2015\)](#). Supports careful choice of evaluation measures, without implying that any single metric is sufficient.

Source boundary. References document mathematical provenance and general concepts. Worked customer examples, numerical parameters, the combined generative law and statements about the current implementation are derived from the project. No source is presented as evidence that these synthetic parameters match a real bank.